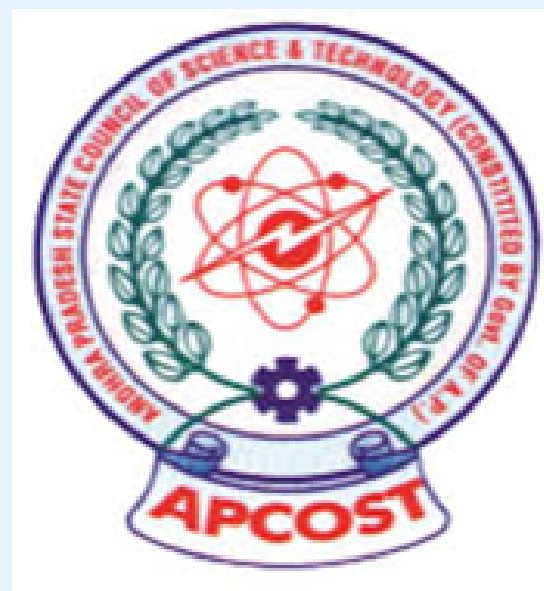


PROCEEDINGS OF INTERNATIONAL CONFERENCE ON ADVANCED MATERIALS, DEVICES AND TECHNOLOGIES (ICAMDT-2023)

6-7 NOVEMBER 2023



Prof.S. Venkatramana Reddy





Dr. S. Venkatramana Reddy is working as Professor in the Department of Physics, S.V.University, Tirupati. He has 22 years' teaching and 28 years research in experience. He has published 130 research papers in various internationally reputed Journals and presented more than 143 papers in various National and International Conferences. Received the Best Paper Award for presentation of the Research Paper entitled "Reducing the effect of ground clutter from wind profiler radar signal using wavelet transforms", in the National Conference. 8 Ph.D. and 4 M.Phil. degrees are awarded under his Supervision. He was Co-ordinator for

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	Applications		
12	Temperature dependent phase formation of Copper Sulphide Nanoparticles by a facile hydrothermal method	Bincy John , Julin Joseph and G. Genifer Silvena	13
13	Preparation and Properties of PVDF/TiO ₂ composite coatings	L. Siva Sankara Reddy, S. Devendra, M. Penchal Reddy , G. Balakrishnaiah and K. Rama Gopal	15
14	Indium Sulfide: A material for luminescence thermometry	Nisha Chandran, R. Jayakrishnan and Rani Abraham	18
15	Elliptical Hollow Core Photonic Crystal Fiber based Mach-Zehnder Interferometer for Analysis of Bacterial Water	Sudipta Kumari Panigrahy, Sukanta Kumar Tripathy	20
16	Electrical, structural and morphological properties Au/n-Ge heterojunctions using V ₂ O ₅ Interfacial layers	G. Henry Thomas, A. Ashok Kumar, S. Kaleemulla, V. Rajagopal Reddy	22
17	An Experimental Study on Inconel 718 Alloy Metal in Different Layer Orientations By Using A Powder Bed Fusion Method	P.Hema N. Balasubramanyam and M. Reddeiah	23
18	Molecular Structure, Spectral, Surface, Electron Density Analysis on Linaloe Oil and Molecular Docking with Human Immunodeficiency Virus-1 Organism (Viral Protein)	Uma Maheswari Chandra Mohan	26
19	Ce ³⁺ /Ho ³⁺ Doped LaPO ₄ Nanomaterials Produced By Hydrothermal Technique For Luminescence Studies using Ethylene Glycol (EG)	Girija Venkateswara Koneru, Sirisha Bandi, Venkata Nagendra Kumar Putta	28
20	Modelling and Analysis of BLDC Motor Control in Simulink for Early Fault Detection and Fast Product Development	Prakhar Mall, Awnish Kumar Tripathi	30
21	Exploration of the Photoluminescence at 1.06 μ m in Near-Infrared and Laser Properties of Zinc Phosphate Glasses Doped with Nd ³⁺ ions for Applications in Solid-State Laser Technology	S. Vidya Sagar, S. Babu , K. Venkata Rao, G. Pullaiah	32
22	Computational investigation on 6-Bromo-1-H-indole 2,3-dione hemihydrate: A combined spectroscopic and DFT analysis	Sunil Raj R and. Lynnette Joseph	34
23	Bio-Physical Investigation of Aluminum Titanate Biomaterials- Osteo Tissue Regeneration	Shanmugapriya B, S. Shailajha, & Sakthi @ Muthulakshmi S	37
24	Calcium Silicate Biocomposites: Effect Of Zirconia on Mechanical Stability and In-Vitro Biological Activity.	Sakthi @ Muthulakshmi S & S. Shailajha	38
25	Growth of 2D CsPbBr ₃ Perovskite thin film by Self-Assembly	Ramesh Chandran B R ., R. Jayakrishnan	39
26	Characterization of Chemically Deposited ZnO: Al Thin Films for CO Gas Sensing Application	Th. Ratanjit Singh, L. Raghumani Singh, A. Nabachandra Singh	44
27	Spectroscopic Investigations of Yb ³⁺ - doped Fluorophosphate Laser Glasses	P. Hari Krishna, K. Krishna Prasad, R. Vijaya, C.K. Jayasankar, N.O. Gopal, P. Sivasanakar Reddy	46
28	Advancements in Remote Temperature Sensing using Photoluminescence: Exploring the Potential of Luminescent Nanomaterials	Suresh Kumar Jakka	48

29	Structural, Optical and Magnetic studies of In/Mn doped ZnO by Combustion Method	S. M. Devasena, K. Sunil Kumar, A. Sharon Smiley, C. Gnanendra, R. P. Vijayalakshmi	49
30	Rare Earth Luminescence Thermometry Based on Fluorescence Intensity Ratio	K. Pavani, A.J. Neves, M.J. Soares, M.P.F. Graça, Suresh Kumar Jakka	50
31	Lead-Free Double Perovskites Structure for Photovoltaic Applications	Kailash Nemade	51
32	Optical, Morphological and structural properties of Rapid Thermal Annealed CoPc thin films	M. Pavani, A. Ashok Kumar, S. Kaleemulla, V. Rajagopal Reddy	52
33	Visible Upconverted and NIR Luminescence Properties of $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ Nanocrystals	N. Ramadevi, R. Praveena, V. Venkatramu, Ch. Basavapoornima, V. Lavín, B.D. Joshi, M.J.Prasad.	53
34	High Performance Bi-Function ZrO_2 : $\text{Er}^{3+}/\text{Yb}^{3+}$ Assisted and CO-Sensitized Dye Sensitized Solar Cells for IoT Applications	Ambapuram Meenakshamma, Parnapalli Muni Mounika, Mudumalla Vikramachari, Mitty Raghavender	54
35	Unveiling A Novel Frontier in Solar Power Conversion Efficiency with a Bi-Facial Tandem Dye Sensitized Solar Cell	Parnapalli Muni Mounika, Ambapuram Meenakshamma, Kalvapalli Susmitha, Ashutosh K Singh, Mudumala Vikramachari, Mitty Raghavender	55
36	Structural and electrical properties of Au/n-GaN schottky junction with a neodymium oxide interlayer	D. Surya Reddy, V. Rajagopal Reddy	56
37	Structural, chemical and electrical properties of Ti/ WO_3 /p-InP heterojunction	S. Asha Jyothi and V. Rajagopal Reddy	57
38	Electrical and Current Transport Properties of Au/CoPc/undoped-InP Metal /Polymer/ Semiconductor (MPS) Structure	A. Usha Rani, K. Ravindranatha Reddy, A. Ashok Kumar, V. Rajagopal Reddy	58
39	Impact of Dy_2O_3 on optical properties of multi-component lithium tetraborate glasses	P. Sai Dinesh, M. Kumar and Y.C. Ratnakaram	59
40	Effects of Photon Field Reduction on Dosimetric Parameters in Radiosurgical Treatment of Single Brain Metastasis.	Shabbir Ahamed, Padma Suvarna	60
41	Effect of PVA Concentration on Structural and Optical Properties of PVA Capped SnS Films Grown By Chemical Bath Deposition	P. Mallika Bramaramba Devi, and K.T. Ramakrishna Reddy	61
42	The Effect of thermal annealing on microstructure and electrical characteristics of Mo/ ZrO_2 /n-Si junction with a high-k ZrO_2 interlayer	V. Manjunath, G. Nagaraju, S. Dastagiri and M. Vani	62
43	Effect of oxygen partial pressure on the physical characteristics of nanocrystalline reactive sputtered Ag-Cu-O thin films	M. Hari Prasad Reddy, S.Venkatramana Reddy, O.Md. Hussain and S.Uthanna	63
44	Facile Solid-State Synthesis and Photo luminescence Properties Studies of CePO_4 : Yb/Ho Doped Nanoparticles	Sirisha Bandi, Phani Raja Kanuparthi, Venkata Nagendra Kumar Putta	64
45	Investigation on the Key Factor for Critical Properties and Rapid Progress Performance of Perovskite Structure based Solarcells	S. Rajalakshmi and P. Venkatachalam	65

Electrical, structural and morphological properties Au/n-Ge heterojunctions using V₂O₅ Interfacial layers

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Abstract

In this paper, we reported the influence of vanadium pent-oxide (V₂O₅) interfacial layer on electrical, structural and morphological properties of Au/V₂O₅/n-Ge Schottky diodes deposited at the substrate temperature 250 °C. The analysis of morphological properties of V₂O₅ thin films on n-Ge surface using AFM and FESEM with EDS reveals the films are homogeneous, continuous and particulate with a high surface coverage. The AFM measurements of V₂O₅ thin films shows a fairly smooth surface with RMS roughness of 3.8 nm measured in the scan area of 5×5 μm². Further, the FESEM with EDS measurements of V₂O₅/n-Ge layers evidenced with V, O and Ge elements signifies the presence of V₂O₅ thin films on n-Ge substrate. The structural properties of V₂O₅ deposited on the n-Ge substrate at the substrate temperature 250 °C were studied from XRD measurements. It shows that the films are highly crystalline in nature and significant diffraction intensity corresponds to V₂O₅ with various crystal orientations are observed. The Electrical properties of the n-Ge heterostructure were studied with and without V₂O₅ thin interlayer. The Schottky barrier parameters such as barrier height and shunt resistance seem to be superior for V₂O₅/n-Ge hetero-junction than compared to without interlayer. The consistence of barrier parameters was observed by evaluating the barrier parameters using other conventional methods such as Cheung's and Norde methods. The current conduction mechanism of V₂O₅/n-Ge hetero-junction signifies SCLC current transport process assisted with traps is found to be dominant at the V₂O₅-Ge interface preferably at the moderate forward bias voltages.

Keywords: V₂O₅/Ge hetero-junction, Electrical properties, Morphological properties, current conduction mechanisms, Structural properties.

Graphical Abstract:

